Towards Sustainable ICTD in Bangladesh: Understanding the Program and Policy Landscape and Its Implications for CSCW and HCI

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Historically, Computer-Supported Cooperative Work (CSCW) and Human-Computer Interaction for Development (HCI4D) researchers in the Global South have advocated for a community-based approach to technology design and development. However, even with this "bottom-up" emphasis, the sustainability and scalability of the resulting innovations remain major challenges, and are poorly understood. To address this gap, we take the case of Bangladesh as a typical Global South context in which development work is carried out by a complex intertwined network of stakeholders across governments, NGOs, donors, and industries. To better understand the current development landscape and its priorities for digital technologies, we conducted interviews with 14 influential decision-makers in Bangladesh who play significant roles in the development of nutrition strategies. Our findings highlight a disconnect between the Bangladesh government's "digital mandate" and the reality of digital innovation practice within the nutrition development sector. Our paper contributes to the debate on factors that affect decision-making processes. We explore the dynamics of diverse actors and institutions who are intended to participate in, but can act as obstacles to sustained bottom-up innovations. Our findings expand understanding of institutional priorities, the dynamics of intermediaries, techno-solutionism, postcolonialism, bureaucracy, competition, and other important topics in CSCW scholarship. We suggest understanding the factors that guide the decision-making process of digital innovation practices in terms of four dimensions: internal, external, vertical, and horizontal. Consequently, we recommend CSCW and HCI researchers become mediators to connect decision-makers and communities and bring their voices in ICT innovations for global development. Finally, we offer recommendations for proactive engagement with decision-making stakeholders, enabling researchers to design community-centered sustainable digital innovations for development.

CCS Concepts: • **Human-centred computing → Human computer interaction (HCI)**.

Additional Key Words and Phrases: Sustainable ICTD, Program and Policy, Development, Implications, HCI, CSCW, Bangladesh

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2573-0142/2022/4-ART126
https://doi.org/10.1145/3512973

ACM Reference Format:
Manika Saha, Delvin Varghese, Tom Bartindale, Shakuntala Haraksingh Thilsted, Syed Ishtiaque Ahmed, and Patrick Olivier. 2022. Towards Sustainable ICTD in Bangladesh: Understanding the Program and Policy Landscape and Its Implications for CSCW and HCI. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW1, Article 126 (April 2022), 31 pages. https://doi.org/10.1145/3512973

1 INTRODUCTION

Over the last two decades in Global South contexts, researchers carrying out Computer-Supported Cooperative Work (CSCW) and Human-Computer Interaction for Development (HCI4D) have argued for the input of local communities, advocating for technology innovation that is culturally appropriate and non-technocratic. They commonly invoke innovation through bottom-up approaches to empower communities within development contexts [35]. However, the inability to sustain a more extended period to scale up technology-based programs and projects remain as absolute constraints in Global South countries [31, 51, 54, 120, 123, 125]. It has also been recognized that for digital innovation to be successful and sustainable, the wider stakeholder landscape needs to be considered alongside local practices and participatory design processes [21]. The importance of understanding the collaborative nature of solution research, design, and use and collaboration with decision-making stakeholders has been emphasized by several researchers in CSCW and HCI [37, 61, 66, 72, 73, 109]. Kumar and Dell [69] go further and have demonstrated the need for a deeper understanding of how global development practitioners use innovation to generate informed practice for successful HCI4D. However, it has remained under-studied why and how the bureaucratically led policymaking process, which often involves embarking on unfounded, overestimated, and expensive technology deployments, is mismatched with the demonstrable value gained from engaging communities on the ground. Further, there is little well-articulated literature on the high-level political contexts and actors in Global South countries, such as the political visions and priorities influencing and driving Information and Communication Technology for Development (ICTD) programs and how projects and policies change as they go through central authorities (from the government to national and international organizations) [124].

To elucidate this space, we take Bangladesh as a representative Global South country. Its technology-centric development programs have been explored extensively in Bangladesh by CSCW and HCI4D researchers [1–4, 16, 61, 90], who have looked at a wide range of marginalized populations and the opportunities and challenges that technology provides for them, taking into account local community perspectives and the broader sociocultural, political, and religious landscape. In this paper, we extend this line of work and focus on nutrition programs in Bangladesh, a priority area of both the Bangladesh government and international development agencies. According to UNICEF, in 2016–17 the Government of Bangladesh spent USD 2.7 billion in nutrition-related interventions, representing around 1% of GDP and about 9% of the national budget [78]. Public health nutrition falls under global health development, one of the major societal issues in ICTD [123]. In Bangladesh, one-quarter of the population (40 million) are severely food-insecure [99]. Achieving Food and Nutrition Security (FNS) is a global priority [88]. FNS is one of the Sustainable Development Goals (SDGs) underpinned by agricultural development through a nutrition-sensitive lens [89, 101]. To achieve FNS, the Bangladesh government has placed emphasis on developing coherent nutrition policies and programs by including policy stakeholder coordination and connection between government, donors, the UN and research organizations, International Non-Governmental Organizations (INGOs), and other decision-making sectors (policies and programs) [98]. Information and Communication Technology (ICT) has been broadly recognized by the government as a medium for the delivery of various services in this space, such as enhancing access to information sharing, communication, and surveillance systems.

Our research aims to understand the high-level decision-making environment in which policies and programs are implemented, and in which digital innovation must embrace bottom-up sustainable impact. We aimed to explore the complex network of decision-making stakeholders to understand their perceptions and insights around current ICT priorities, collaboration among stakeholders, and practices for program development and implementation. To accomplish this, we conducted 14 interviews with influential high-level Bangladeshi policy decision-makers, central government officials, senior NGO staff, and UN officials, who are actively involved in shaping Bangladesh's current nutrition policy and program strategy. These included influential high-level Bangladeshi policy decision-makers, central government officials, senior NGO staff, and UN officials. We also applied these insights to our broader query on the actual practice of the bottom-up approach in ICT-related program formulation and implementation.

This work makes several contributions to the computer-supported cooperative work (CSCW), and Human-computer interaction (HCI) and information and communication technology (ICT) for development literature. First, we found that there is a disconnect between Bangladesh's digital mandate and actual development at the ground level, which is greatly influenced by top-level decision-makers such as donors' interests and political priorities in the public health nutrition sector in Bangladesh. Moreover, decision-makers have different views on digital innovation, which emerge from decision-makers' predilection for "solutionism" and from unmet expectations. Second, we also found that collaboration among diverse institutional stakeholder across nutrition agencies is disjointed, which creates power dynamics, political tensions, and unintentional competition between decision-makers. Thus, we seek to understand what drives decision-makers' priorities in Global South contexts that inform uptake of certain innovations over others and how this affects the marginalized population's real prospects for development [124]. In parallel, we discuss the complexity of the decision-making environment, with our findings revealing important factors that impede sustainable innovation of bottom-up approaches within development contexts. Our third and key contribution therefore suggests a way to understand this complex context.

We suggest understanding the factors that guide the decision-making process in terms of four dimensions-internal, external, vertical, and horizontal-toward better understanding decision-making processes and to suggest a way forward for CSCW and HCI researchers as mediators to bridge the gap between decision-makers and communities for effective innovations. Finally, we highlight several recommendations, challenges and questions for CSCW and HCI4D researchers that need to be considered in future work and research in Global South contexts.

2 RELATED WORK

2.1 Digital Technology and HCI for Sustainable Development

CSCW, HCI, and related fields have long demonstrated their potential to influence international development where ICT is involved, and thus the intersection between HCI and ICT for Development is often at tailoring technologies for marginalized populations [60, 117, 127]. The review on HCI4D by Dell and Kumar noted that, *"nearly half of the papers (128) target 'ground-level users' to respond to their specific needs and challenges"* [35]. With this, we see a significant amount of HCI4D work has been conducted in the areas of sharing agricultural communication and information with farmers [43, 94], health [68, 92] and nutrition for women [23, 110], facilitating field tasks with health workers [100], educational technology [121], monitoring and evaluation (M&E) purposes for NGOs [11, 74], and safety and security goals for women [3, 5]. Furthermore, researchers have also shown that there is a big push in the Global South to transfer technologies from the West and use them in international development contexts, copying the western practice of innovation [59], which is compounded by the UN and other NGOs advocating for the innovation and use of digital technology towards achieving all SDGs [89]. While there is a definite upward trend of using ICT in most development programs and projects, an inability to sustain such innovation for an extended period after the initial research is conducted, so as to be able to scale up technology-based programs and projects, remains an absolute constraint [8, 31, 51, 54, 123, 124]. In their review, Avgerou and Walsham [9] stated that *"successful examples of computerization can be found ... but frustrating stories of systems which failed ... are more frequent."*

Understanding local contexts, including appropriate technology design and selection and social, cultural, economic, and political factors, is highlighted by many researchers on the developing world as important in addressing sustainable ICTD [8, 51, 53, 120, 123]. James C. Scott [105] showed, in relation to his large program failure cases, that successful program design needs to value local and practical knowledge for implementation. Anokwa *et al.* [6] also emphasized the understanding of challenges and the importance of collaboration at the local levels. Conversely, the unintentional practice of ICTD researchers conducting short-term visits to complete their research to make publications and recommendations without any meaningful impact is also highlighted as an ethical concern and a reason behind the lack of sustainable impact of ICTD in Global South contexts [32, 33]. In response, Dearden and Tucker [33] suggested that *"actors in the broader field of ICTD implementation (i.e., action by NGOs, aid agencies, and other organizations innovating and applying technology to development challenges) should examine how resources are used and how decisions are taken so that systemic and sustainable capacity building is prioritized."*

The importance of understanding the high-level decision-making context has also been well recognized and highlighted by the broader HCI community [12, 57, 72, 73]. While many studies focus on the bottom-level local context of program implementation [6, 31, 120] and note the importance of understanding political analysis [123], there has been sparse reporting on the high-level context in which policy and programs are formed. Heeks emphasized that the high rate of ICTD program failure arises from the "design-reality gaps" that arise from a lack of understanding of decision-makers' and program designers' contexts [53]. Overall, sufficient insights have not yet been articulated on the high-level political contexts and actors, such as the political visions and priorities influencing and driving ICTD programs and how projects and policies change as they go through a central authority (from the government to national and international organizations, including donors) [124]. We intend to contribute to this discussion by presenting the nature and roots of these gaps in the public health nutrition context in Bangladesh.

2.2 CSCW and HCI in Program and Policy Decision-Making Contexts

CSCW and HCI for program and policy research have a historical perspective informed by the work of social informatics researchers such as Kling, King, and Scacchi in the early '80s [65–67]. Kling provided rich insights on the "cooperative" and "collaborative" nature of social computing and the need to tie design, use, and technological impact together among stakeholders [66]. CSCW, HCI and ICT have historically had a relationship with participatory design (PD) and innovation-related programs and policymaking by specific political concerns [62, 65–67, 122]. Jackson *et al.* [62] recognized that there is a disconnection among research, design, and practice in the field of CSCW and argued for a holistic orientation, proposing an interconnected "knot model" (among design, use, and collaboration with organizations within public policy systems) with a focus on policy, practice, and design. The role of policy design was also investigated by Jackson and colleagues, [63] who emphasized the need for collaboration and mutual learning between scientific study and policy to successfully navigate design, infrastructure, and organizational collaboration complexities. They highlighted that traditional CSCW is more focused on "implication for design" than "implication for policy" in regard to ambitions of long-term social impact. Similarly, Freeman *et al.* (2017) [42] pointed out the scarcity of empirical studies between design and policy contexts in HCI and CSCW.

Their empirical study offers a critical viewpoint by showing how non-western urban computing towards a smart city involves citizen design and public design policy to achieve socio-technical systems in which policies and politics are shaped. Towards a robust and democratic technological design process, they urge that a broader and more diverse group of voices be heard and engaged in city life design processes. Kumar and Dell also highlighted the need to understand the challenges to established "informed practice" for HCI4D [69] in adapting technologies by global development practitioners. In a fieldwork study of mobile-based health care intervention for a marginalized community in Uganda, Ho *et al.* argued that a sustainable approach to innovation needs not only a design-based approach for the users, but also needs to involve decision-making stakeholders so as to consider all stakeholders' needs, priorities, and conflicting issues [56]. The importance of understanding the role of HCI researchers in the complex array of national and other relevant actors involved in decision-making processes has also been highlighted by Thomas [116].

The complex multi-sectoral nature of high-level decision making and the institutions that form these decision networks can be further understood through the concept of institutioning. Institutioning refers to working within diverse institutions [36], and is a useful frame for understanding the complexity within community-based design processes, particularly when conducting a program with a participatory design (PD) [114, 115]. These scenarios are characterized as situations in which the process is constantly moving horizontally and vertically between and within diverse micro-meso and macro levels [58] and call for attention to the significant relationships between policy, institutions, and meta-cultural frames [58, 115]. PD approaches in social computing are figured by a diverse group of institutions, and need to be understood in terms of the meso-and macro-level institutions that PD is dependent on and informed by [58]. Dearden and colleagues [33] also acknowledged the need to understand the institutioning involved in critical computing issues. In the debate on HCI and social innovation, they also contribute to the insufficiently studied issue of HCI designers' lack of attention past the research context to how innovations become distributed and adopted for social goods. Political issues surrounding technology, such as how decisions are made for technology adoption and scale-up in programs may be understood through the complexities of institutioning and the actors at the decision-making level [114, 115], where these institutions and their individuals are broadly influenced by their own objectives and priorities, yet have to translate initiatives between designers and communities [25, 26].

Actors who interact with both designer and users are considered "intermediaries." They create opportunities to share, use, and implement technologies and technological innovations toward social innovation [25, 111]. HCI researchers need to understand the relational dynamics of intermediaries in social innovation processes. Through configuring the role of intermediation and understanding how an artefact is shaped, embraced, and used in the real world, PD can improve HCI's approach to effective social innovations. Huybrechts *et al.* [58] also argued that besides doing micro-political work, PD researchers need to expand their boundaries *"for a more critical acknowledgement of the ways in which PD processes are inscribed in, and dependent on, institutional actors."* Studies have noted that for better scaling of PD practice toward social innovation, we need to understand the "institutional frames" in which institutions' rules, agendas, and priorities work [58, 76]. Various studies in the literature have highlighted gaps in understanding how innovations are shaped by intermediaries as actors; institutions; institutional norms and cultures; and actors' objectives, interests, practices, and political agendas [25, 26, 33, 34, 58, 114, 115].

Within this diverse high-level decision-making institutional context, the literature broadly highlights some common challenges such as bureaucracy and hierarchical power dynamics. A recent paper in the interventionist HCI4D and ICTD context presented the bureaucratic challenges among government officials that resulted from moving from a paper-based system to an e-governance system [80]. Another example, Dow *et al.* [37] worked with top-down stakeholders for collaborative design and development in grassroots programs and highlighted inflexible bureaucratic, hierarchical challenges in engaging policymakers in citizen platforms' policy-led designs. The complexities of the top-down decision-making context are thus well recognized, as reflected by Lazar's statement that *"taken broadly, public policy includes not only laws, regulations, enforcement actions, lawsuits, and court actions, but also human rights treaties, international technical standards, non-governmental organizations, and multinational organizations."* [73]. There is currently a limited discussion in HCI, emphasizing the need to understand actors' participation in high-level decision-making environments (who initiates, who benefits, and how these are directed) [124].

A study by Spaa *et al.* [109] suggested aiming for more engagement with these policymakers in three ways: i) finding ways to translate evidence-based policy-making; ii) identifying the role of HCI researchers in engaging in design-led approaches in a decision-making space; and iii) exploring diverse stakeholder collaborations at the decision-making level to understand their priorities and uses of technologies that inform policies. A reflection work by Nathan [87] shared experiences working with United Nations International Criminal Tribunal in Rwanda, and suggested five levels of policy aspects HCI researchers should explore and consider: i) government (state level), ii) global (collaboration with different countries), iii) institutional (individual organizations' level regulations), iv) working-group norms and expectations, and v) HCI researchers' own principles, expectations, and policies. Moreover, in the Global South, this high-level decision-making context is especially complicated, as in the post-colonial era the international development context works with diverse sectors with multiple stakeholders from governments, the UN, INGOs, and donor organizations to develop and implement programs at the ground level. According to Walsham [124], for *"South-South and triangular cooperation, political analysis is needed on who is doing the driving of such projects and who benefits. There is a danger that the driving is coming from the rich North and that the Global South plays a subordinate role."*. He discusses this socio-political background as a "black box" in which there is a lack of understanding of this decision-making context in countries in the Global South [123].

2.3 Digitization in International Development

In the current democratic digital age, many prominent scholars in information technology and development studies have explored the political motivations and complexities of governance, institutions, and technology adoptions in global development. Many studies have investigated the common political tendency of governments in the Global South to push for innovations for all types of development. Gurumurthy points out that the tendency of this push for digitization extends power dynamics and contestations in top-down contexts [46–48]. Likewise, Maitet *et al.* critiques this promotion of digitization by governments as tending to widen the digital divide between the poor and the rich [77]. Through a meta-analysis of various digital contexts, Burns [22] highlights this promotion of digital innovation in light of governance and democracy-related political issues. He critiques decision-making stakeholder collaboration by governments, NGOs, INGOs, and donor organizations working towards international program development and implementation through digital innovation as a process driven by capitalism, resulting in wealth inequality and ignoring peoples' needs [22]. Similarly, the renowned economists Easterly [38] and Moyo [40] have criticized donor institutions as enlarging national-level government bureaucracies, preserving bad governance, and widening the inequity between rich and poor. They highlighted how factors of social and cultural capital impact participatory development in both top-down and bottom-up approaches to sustainable development in the Global South [112].

Extensive work in a variety of e-governance sectors has been conducted by Bhatnagar and colleagues, primarily in India. They highlighted many potentials of e-government and also noted how inefficient and corrupt bureaucracy and overall poor-quality governance restrict sustainable innovations in developments in the Global South [13, 14, 24]. Toyoma broadly working in the interaction of ICTD and HCID, argued that digital innovations may amplify human development in a broad set of global development contexts, where institutions and responsible individuals work with optimistic and good intentions [118]. However, it has not been well-explored how multi-sectoral decision-makers, including national governments, INGOs, UN organizations, and donors, work, perceive, and interact in a very specific development sector (such as the public health nutrition development context in Bangladesh) towards digital innovation and their implications for CSCW and HCI practitioners for sustainable development.

The literature on CSCW, HCI, ICTD, and development studies described above help us to understand the broader stakeholder context. They provide a foundation for exploring the current higher-level decision making contexts in the public health nutrition sector in Bangladesh. To sum up, the diverse collaborative decision-making settings in development contexts raises several concerns regarding the voices that count in this decision-making process and the factors that influence the incorporation of new programs, research, and interventions into developmental programs and policies. The political priority for global initiatives is often understood by interactions among four factors: i) the power of involved actors, ii) the notions they use to illustrate specific problems such as health-related issues, iii) the political contexts in which they act, and iv) the characteristics of the issue itself [106]. We posit that understanding this complex high-level decision-making context in developing countries as crucial; such contexts include the decision-makers' priorities, objectives, and mandates, the factors that create tensions, competition, and challenges among multiple institutional stakeholders in the decision-making process, and the current ICT practices in program development and implementation that reflect decision-makers views and perceptions. These issues are essential for CSCW scholarship, as we believe this high-level decision-making context directly impacts the bottom-level initiatives that aim to make a real and positive impact on marginalized people's lives.

3 CONTEXT: NUTRITION DEVELOPMENT IN BANGLADESH

Bangladesh has endured and survived many wars and poverty-related challenges in the last century; the narrative around nutrition development is interlinked with this history. Pre-1947, Bangladesh was part of the British Raj and was the epicenter of the Bengal famine (1943), which is said to have taken the lives of 2 million people (3.3% of the population) [45]. In 1947, modern-day Bangladesh was under Pakistani rule, a state of affairs that was to change during the bloody liberation struggle of 1971. Less than a generation after the specter of the Bengal famine, the Independence struggle saw the people of Bangladesh re-living mass starvation, countless deaths, and economic disasters. Also, being located on the Ganges delta, the largest river delta in the world, Bangladesh has consistently experienced heavy annual floods and the resultant devastation to people, livestock, and crops.

Taking this history into account, it is perhaps not surprising that nutrition and public health are enshrined in Bangladesh's constitution, where it is written that *"the State shall regard the raising of the level of nutrition and improvement of public health as among its primary duties."*[1] From the beginning, international development has been part of Bangladesh's national development narrative. Under the Ministry of Health and Family Welfare's Bangladesh National Nutrition Council (BNNC), the first nutrition policy, the National Plan of Action for Nutrition (NPAN), was initiated in 1980 funded primarily by the World Bank [113]. Though BNNC was supposed to work autonomously, there is a lack of political commitment by the government and coordination between the government and development partners [79]. BNNC was virtually ineffective for many years, until the formulation of a second nutrition policy in 2015, NPAN 2 (2016–2025). Headed by the

[1] Article 18(1), Bangladesh Constitution

Bangladesh Prime Minister, emphasis was explicitly placed on the revitalization of BNNC and strengthening of the collaboration between different stakeholders [107].

3.1 The Actors - Stakeholder Collaboration

While nutrition policy agendas are discussed and put into law by decision-makers in the capital, Dhaka city, projects to support the policies are undersigned by largely international donors (such as the World Bank and USAID). As a result, a strong and direct influence of global movements is visible in nutrition programs and policy development. For instance, the Millennium Development Goals (MDGs) of 2000 introduced specifically-targeted national nutritional surveillance system indicators. Various studies have highlighted the importance of integrating all relevant sectoral stakeholders for evidence-based nutrition policy formulation and implementation more coherently and collaboratively [15, 17, 49, 50, 103]. In 2014, the Second International Conference on Nutrition (ICN2) declared, *"it is well-established that nutrition objectives can only be achieved through a multi-sectoral response. This requires different sectors and stakeholders working together more coherently and collaboratively to address malnutrition in all its forms"* [41].

Bangladesh is also a member of the Scaling Up for Nutrition (SUN) movement, which is strengthened by a political concern in nutrition development among government and development partners to coordinate and align action through programs to implement policies [91]. Moreover, SDGs have recently increased attention to nutrition development with a similar and comprehensive goal to address "malnutrition in all its forms" [88, 101]. Bangladesh's national nutrition decision-makers have been working with an extensive network of local, national, and international organizations: (1) 24 government ministries (including the Ministry of Health, Ministry of Agriculture, Ministry of Food, Ministry of ICT, and Ministry of Finance); (2) an extensive network of International NGOs (including Save the Children[2], GAIN[3], Concern[4], and Nutrition International[5]); (3) United Nations agencies (including UNICEF[6], the Food and Agriculture Organization[7], the World Health Organization[8], and the World Food Program[9]); (4) Research organizations (including icddr,b[10] and CGIAR[11]); and (5) Donors (including WorldBank[12], USAID[13], DfID[14], AusAID[15], and BMGF[16]). As might be expected, working with a large network of stakeholders on nutrition policy and governance creates a complex policymaking environment, in which the political interests of current government and non-government stakeholders are intricately intertwined [113].

3.2 ICT in Nutrition Policies and Programs

Acknowledging the need for coordinated action for nutrition development, the UN and other international organizations have advocated for digital technology to support program and policy implementation. The nutrition development sector has broadly focused on two types of interventions: nutrition-specific interventions, which are preventive initiatives such as supplementation and immunization, and nutrition-sensitive intervention, which is broadly focused on addressing underlying causes of malnutrition by enhancing food-security via improving agriculture food systems, women's empowerment, water, hygiene, and sanitation, among other interventions [103]. Nutrition has been mainstreamed within existing health and agriculture systems in Bangladesh, where digital technologies have been broadly recognized as a medium for service delivery. Service delivery is an umbrella where digital technologies have been touted to circumvent challenges in

[2]https://www.savethechildren.net/ [3]https://www.gainhealth.org/homepage [4]https://www.concern.net/
[5]https://www.nutritionintl.org/ [6]ttps://www.unicef.org/ [7]http://www.fao.org/home/en/ [8]https://www.who.int/
[9]ttps://www.wfp.org/ [10]https://www.icddrb.org/ [11]https://www.cgiar.org/ [12]https://www.worldbank.org/
[13]https://www.usaid.gov/ [14]https://www.gov.uk/government/organisations/department-for-international-development
[15]https://www.preventionweb.net/organizations/1172 [16]https://www.gatesfoundation.org/

information access, communication, connection, and other services, including market linkages and surveillance systems.

Finally, Bangladesh has not been immune to the sweeping changes introduced in development practice by the proliferation of mass communication technologies in the last few decades. These changes have caught many development stakeholders blindsided, and many have struggled to effectively engage with technologies within projects in the development sector. There is an interesting paradox here when we look at the national setting. Consider the Digital Bangladesh slogan, which was championed during the 2008 election campaign by the Awami League government, who promised that the party would *"undertake a modern philosophy of effective and useful use of technology in terms of implementing the promises in education, health, job placement, poverty reduction, among others"* [44]. This political mandate effectively embedded digital innovation as crucial to the country's developmental progress. This agenda is also evident in new initiatives such as the "Access to Information (A2i)" flagship program of the Bangladesh government [39], *"heralding a mindset change within Government to embrace ICTs as a powerful enabler for the nation's socio-economic transformation"* [52].

4 METHODS

We conducted a qualitative study with nutrition policy and program makers for whom innovation through digital technology is a core part of national-level nutrition program development and implementation in Bangladesh.

4.1 Recruitment

We conducted semi-structured interviews (face-to-face) with 14 experienced decision-makers from within Bangladesh's nutrition development sector. They were recruited through direct email contact and recommendations from our acquaintances in the sector. Potential participants were fully briefed about the research scope by email and data collection only proceeded once consent was given. The ethics review board of the authors' institution granted full ethical approval.

4.2 Participants

The decision-makers represented various organizations. They consisted of (i) policymakers: government officials who play a role in creating policies and leading nutrition government agencies and (ii) policy-influencers: leaders of prominent national and international NGOs, research organizations and the UN agencies who are consulted when policies are created. These decision-makers, whose demographic information is shown in Table 1, are experienced practitioners who have a combined experience of over 277 years working in the nutrition sector and are all based in Dhaka, Bangladesh. All of them are involved in different nutrition programs implemented in Bangladesh. They are all well-educated; none of their educational backgrounds is less than a Masters' degree.

In addition, most participants assist relatively high-level positions at the organizations they work for, such as director, CEO, advisor, and scientists (senior researchers), who are generally involved in high-level decision-making in nutrition programs and policy in Bangladesh, including determining their organizations' program planning, managing, and coordinating. Our interview sought a broader understanding of these high-level decision-making contexts from the perspective of these stakeholders, including their priorities, visions, perceptions, and expectations of innovation for development. We therefore believe that the participants we recruited were well-qualified for our study. Their experience and high-level positions in the development sectors certainly helped us capture varied and rich data on our targeted topics.

Table 1. Our participants are experienced and influential policymakers and policy-influencers in Bangladesh's nutrition development arena.

Code	Experience	Affiliation	Education	Gender
P1	~15 years	Ministry of Agriculture	MSc	Male
P2	~15 years	Ministry of Health	PhD	Male
P3	~25 years	Ministry of Livestock	PhD	Male
P4	~20 years	Ministry of Food	MSc	Male
P5	~20 years	Ministry of Fisheries	MSc	Male
P6	~15 years	UN agency	MSc	Male
P7	~30 years	UN agency	PhD	Male
P8	~15 years	UN agency	MSc	Female
P9	~30 years	UN agency	PhD	Female
P10	~25 years	ICTD Think-tank	MSc	Male
P11	~12 years	Nutrition Think-tank	PhD	Female
P12	~20 years	Nutrition Think-tank	PhD	Female
P13	~15 years	Nutrition Think-tank	MSc	Female
P14	~20 years	International NGO	PhD	Female

4.3 Interviews and Author Experience

The first author of the present study, a native Bengali speaker, carried out these interviews in Bengali during March and April of 2019 in Dhaka. Each of these individual interviews were 60 to 90 minutes long. We stopped the interviews when we reached a theoretical saturation [93], at which further interviews started adding very little new information. Our research is reflective of broader ICTD realities in the Bangladeshi context. Two of the authors were born and raised in Bangladesh and are very familiar with the local context. In addition, the first author is an experienced public health nutritionist who has five years' work experience in various development projects (working with a range of people from those in high-level program- and policy-making to those at in the ground-level community) in Bangladesh. The first author is currently exploring broader research under the HCI4D domain with a practitioner point of view. The second Bangladeshi author has more than 12 years of experience conducting ICTD and HCI4D research in Bangladesh. Another author is a public health nutrition expert who has more than 40 years of work experience in Global South settings. All of the other authors are well experienced in international development and the HCI field and in qualitative research.

4.4 Semi-Structured Interview Schedule and Interview Style

This research aims to illuminate high-level decision-makers' priorities, views, perceptions, and expectations of innovation for innovation-centered program development and implementation. We critically explore the main drivers and how power dynamics work among diverse stakeholders as they incorporate ICT solutions in nutrition program development and implementation. From this critical standpoint, we have also tried to determine the actual role of the bottom-up approach in high-level decision-making in Bangladesh. We developed a semi-structured interview (SSI) schedule plan with the following high-level categories: i) collecting demographic and background information, ii) understanding decision-makers' motivations in using ICT during program development and implementation, iii) understanding decision-makers' priorities in using ICT during program development and implementation, iv) understanding decision-makers' perceptions and expectations of innovation, and v) understanding existing power dynamics decision-making among diverse stakeholder. After recruitment (details in Section 4.1), the first author visited participants' offices according to their given convenient schedule. We followed our high-level themes based on our SSI, although the interview style was more open and informal. Interviews were conducted according to a conversational approach in which participants had some latitude to share from their broader practical experiences in public health nutrition decision-making settings in Bangladesh.

4.5 Analytical Approach

We applied the thematic analysis method to analyze the interview data. All interviews were audio-recorded and transcribed in Bengali. The first author, a fluent speaker of Bangla and English, individually read the transcripts several times and then partially translated them into English. The anonymized data were subsequently analyzed using open coding, following the methodological lessons of Thematic Analysis (TA) by Braun and Clarke [20]. The analysis began by iterating through the data several times and allowing codes to emerge. Examples of first-round codes included influence of global agendas, donors' interests, the meaning of "innovation," lessons from neighboring countries, stakeholder collaboration, and innovation for youth. After the first round of coding, the authors discussed the generated codes and themes, and the first author presented excerpts from the transcript to justify each theme. Following the discussion, the codes and themes were refined. This process was then repeated twice, and we iteratively refined the codes into higher-level themes and categories that represent the findings described below. This process enabled us to analyze the interviews in the source language (Bengali) and capture the rich meanings in the discussions. It is worth noting that the authors' experience as practitioners in the social-construction approach [27] also shaped the interpretation of the data in a meaningful way.

5 FINDINGS

From our analysis of the interviews, we report on three themes that emerged that may inhibit a bottom-up approach to sustainable ICT program development in Bangladesh. In the following sections, we first focus on 1) the factors that shape innovation and higher-level stakeholder (governments and donors) priorities for ICTD programs, 2) an evidence-based approach to ICT programs, and 3) competition and power dynamics. We unpack the first factor into four areas: i) how digital mandates and donor interests shape ICTD programs, ii) how decision-makers' perceptions of innovation are shaped by their accountability to higher-level stakeholders, iii) pressure and competition to create more innovations during implementation, and iv) linking young people to technological adoption results by default, ignoring other groups in the population. We then discuss how the evidence-based approach overlooks the bottom-up approach to ICTD in nutrition programs. Finally, we report on competitions and power dynamics among decision-making stakeholders, specifically between those at similar levels in the power hierarchy.

5.1 Innovation and Higher-Level Priorities for ICT-Related Program Development and Implementation

5.1.1 Digital Mandates for Innovation and Donors' Interest in ICT. Innovation is increasingly becoming *"mandatory ... because the government is trying to become a middle-income country by 2025"* [P9]. To achieve this vision, the Bangladeshi government has the political manifesto of establishing "Digital Bangladesh" to enact ICTs as the mainstay of all development. The Bangladeshi government's vision of progress and global recognition plays a significant role in the way its top decision-makers pursue scalable development interventions underpinned by digital technologies. Our interviewees commented on how this government-led political mandate drives top-level decision-makers to actively consider technologies within portfolios spread across different government sectors. For example,*"Bangladesh is doing a great job in digital health innovation. We have won international recognition for this. Of course, it is a political mandate, but all sectors need technologies"* [P2]. All of our interviewees pointed out that this political mandate drives top-level decision-makers to keep ICT as a core factor in all development initiatives. Indeed, the centrality of technology to the Bangladeshi government's modernization narrative is found in the urgency *"to incorporate technology everywhere; it's a political matter"* [P11].

This techno-progressivist vision, in which the nation's growth is intricately linked to its ability to utilize digital innovation, is further cemented in the development discourse due to its partial alignment with the United Nations' SDGs. The government's high priority of achieving the SDGs has been reflected in their recent initiative of introducing a digital "SDG Tracker"[17] to monitor their national development progress, promote accountability, and take responsibility for *"updating others about progress on global initiatives"* [P9]. Our participants shared that they are responsible for regular reporting to the UN and other donor organizations on how their development initiatives have kept pace with expected progress. One of our UN participants (a policy influencer) also commented: *"Bangladesh is responsible for updating their SDG indicators to the global bodies of which they are members in order to update them on details on their progress"*[P7]. For many NGOs, keeping up with these radical changes requires substantial alteration to their existing workflows: *"we had to report some indicators, and previously it was all paper-based, but very recently we have developed a real-time digital monitoring system. This is quite an achievement already!"* [P2]. Interestingly, the interviewees opined that a desire for innovation is not only a priority for high-level government officials and political manifestos, but it is also shared by NGO field workers and implementers. Hence, leveraging specific digital technologies has been seen as an anchor to achieve these national and international mandates and may directly impact ICT program development.

Moreover, international donors are another set of stakeholders that shape innovation and ICT program initiatives in Bangladesh. Participants indicated that national and global agendas (such as SDGs) and donors' interest in specific initiatives influence their priorities in interventions, research, and program development. As in other development sectors, our participants reinforced the fact that there are vast funding opportunities available to support public health nutrition development initiatives in Bangladesh. They also supported earlier research findings that in this bureaucratic context, ICT for development programs are run according to what the *donor* wants and how respected organizations can respond to them accordingly. Our interviewees confirmed a high tendency among the donors to seek technology innovations in programs and projects, which is important because response to innovation is positively related to getting funds for projects. Participants also pointed out, however, that donor interest in program formulation creates competition among the recipient organizations and raises tensions in discussions on who gets funding to implement their programs. Actual community needs and project feasibility within local constraints are often not considered a priority within processes. This sentiment was reflected by an interviewee who works in an INGO:*"When some of our big donor groups asked us about telemedicine ... all of us became interested in telemedicine-based interventions and rushed to scale it up as well without attention to the local capacity and their ultimate advancement"* [P12].

In brief, digital innovations are seen as a core means of achieving national political manifestos and global agendas such as fulfilling SDGs for health and nutrition in Bangladesh. Moreover, donors' interest in and funding priorities for specific digital health innovations also drive respondent organizations to incorporate ICT-related interventions in health and nutrition programs. Thus, the higher priorities of political agendas of the government, donors, and INGOs interested in specific technology initiatives often lead to limited consideration of needs and priorities at the bottom level for those programs initiated. This limited focus on local community contexts creates barriers to directly addressing bottom-up needs, challenges, and priorities and results in a conflicted vision of sustainable development.

[17]https://a2i.gov.bd/sdg-tracker/

5.1.2 A Tendency to Create More Technologies. Digital technologies (DTs) have provided an effective way to manage nutrition development programs in Bangladesh. Our participants recognized DTs as a solution-focused approach, emphasizing technology-based interventions for effective program implementation. These technologies would not only be for knowledge dissemination; but organizations could also use them to effectively communicate behavior change protocols and monitor health indicators across population groups. Our participants described a widespread desire for more such technology-enabled innovation solutions. According to a UNICEF official, *"there is a huge appetite for technology solutions in our field"* [P8] and *"we desperately need technology-enabled solutions"* [P8].

Six participants strongly believed that innovations equates to more social-media-based platforms and apps for service delivery purposes such as learning, training, sharing information, and business processes. A participant from an UN organization noted that *"We could use ICT to develop more apps like Facebook and more mobile apps"* [P8]. An INGO participant echoed another comment on the importance of more social media: *"more new initiatives around Facebook, messaging, and mobile apps will help knowledge dissemination"* [P2]. However, a common tendency among stakeholders was noted: despite working within the same domain, NGOs and public agencies often built their bespoke applications and web platforms with little collaboration or knowledge shared with each other. Our participants recalled several NGOs and government departments who built apps that carried out very similar activities. For instance, one participant specified two such overlapping apps that were made independently of each other: *"technology-based apps for nutrition education, such as the 'farmers window' app and 'mojay mojay pusti shika' [learning about nutrition through fun]"* [P2]. Another participant described an app pioneered by the government: *'pusti app' [nutrition app] is used for giving accurate information about nutrition, methods for cooking, calorie information based on measuring BMI, etc."* [P1]. Notably, they did not mention using or adapting each other's work in their program implementation, mentioning only their brand-new initiatives and the importance of developing more apps and technologies.

This tendency of each agency to birth their own apps and platforms independently of what the others do was considered inherently problematic. An NGO member opined that *"people always want to start up something new. That is a big challenge. If you start something new and others are also starting new, then what will happen to the existing facilities?"* [P10]. Arguably, the repetitive and duplicative nature of digital initiatives may reasonably be seen as a waste of precious resources, and ironically, innovation throws the spotlight on the inherently weak collaboration between stakeholders. This tendency to build and use their own new technology rather than rely on each other's work can be blamed on competition to claim their own contributions as new initiatives. Participants put forward their own reasons for why this pattern exists in Bangladesh. For one, there are political factors behind the intention to create new technology-based initiatives in nutrition programs. The government of Bangladesh wants to see increased technology usage in the sectors within which our participants work (in line with the Digital Bangladesh agenda): *"it's a political matter; you need to show something innovative and a new approach to handle the challenges because the government and donors want to see it"* [P12]. As novelty is equated with innovation, an unfortunate side-effect of this pressure is the proliferation of new technologies that do not all end up being used effectively. One research officer described how they succumbed to the pressure to innovate and created a new bespoke platform for providing gender-sensitive nutritional information: *"We developed it for the women farmers, but that technology did not work at the field level, as most of them do not know how to use it"* [P11]. For our participants, this realization occurred through the challenges they experienced in their careers working in the Bangladesh nutrition sector, which continue to be a significant factor in program development.

Our findings suggest that there is a tendency among the diverse decision-makers to create their own technologies for nutrition programs, with little multi-stakeholder collaboration. Competition, lack of contextual understanding of digital innovations for communities, and political pressure to create new technologies can too often result in duplication, repetitiveness, and failures of ICTs in public health nutrition. In a rush to create new technologies, the target populations' characteristics, requirements, and capacities are not fully considered.

5.1.3 Practices, Perceptions and, Expectations. Like other development sectors, there is a wide range of funding opportunities available to support nutrition development in Bangladesh. As mentioned above in section 5.1.1 our interviewees confirmed a high desire among donors for technology innovation in programs and projects. Understandably, organizations' response to innovation is often closely related to obtaining funds for projects for such donors. However, our participants also highlighted that decision-makers seek innovations in health, nutrition, and agri-food systems to improve the quality of interventions for better outcomes. For instance, according to participants, they seek innovations related to new approaches and technologies needed in health and nutrition promotional programs such as awareness building, clinical services such as regular nutrition monitoring systems, and other service deliveries such as improving the nutritional quality of food and food security (availability, accessibility, utilization and stabilization of nutritious food for all).

However, from the policymakers' perspective, being involved in NGO programs, interventions, and research initiatives often allows them to learn about new approaches that can be supported to deliver current government priorities (without needing to devise new programs from scratch). Moreover, some NGO workers acknowledged the potential ethical nuance involved in such approaches, noting that sometimes these collaborations can be tokenistic: *"it becomes about searching for a place to insert the policymaker's name into the latest innovative program... because when a new policymaker comes, the first thing they search for is new initiatives with innovations...it's more political than ethical"* [P12]. In these cases, NGOs' quest for national recognition and promotion of their work to larger populations may lead to ethical compromises. Searching for innovation for the betterment of the people is crucial, but our findings pointed out a common political tendency in which innovation is mostly driven by striving for continued funding opportunities and is adopted as a strategy of policymakers to achieve recognition in national development. This again raises the question of how bottom-up approaches of innovation can be valued from a high-level perspective.

Participants also shared their perception of technology innovation. They (policy and program-makers) believe innovation can make their jobs easier and more effective. For our participants, innovation is defined primarily in terms of an increase in worker productivity: *"Innovation means using technologies and making things easy to perform"* [P12]. Innovation through technology within the nutrition sector is expected to enable them to maximize the productivity of both nutrition strategists and field workers: *"Innovation means something that makes everyone's work easier and also helps you to make your best effort by giving small input but for better output"* [P2]. Innovation is also perceived as a direct route to scalability. It includes new approaches that enable mass communication and effective dissemination of information: *"innovative approaches are handy and make it easy to ... share knowledge ... in a convenient way to the larger population"* [P2]. One concrete expression of this form of innovation is described by [P12], an NGO-based senior researcher experimenting with cutting-edge approaches to reducing the amount of worker effort and expertise needed in gathering malnutrition data: *"To assess malnutrition status among children, we are working with a university to build a system where you can measure nutrition status just by taking a picture"* [P12].

Additionally, while our participants shared the need for innovation to make their job easier and to find better interventions for effective programs, there has been a common tendency that directly relates their accountability to their higher-level management and donors. All of the participants noted the need for more innovation in current monitoring and evaluation (M&E) processes. Moreover, M&E is a traditional accountability approach to high-level management and donors to update and show a program's signs of progress. It is a core component of every program-development management system, needed to assess the program's progress, impacts, and challenges. NGO interviewees often shared their aspirations for technologies that would bring radically new solutions to current challenges in M&E, for instance, the challenge of real-time access to accurate population data for practitioners wherever they are based, whether in the rural field or in NGO headquarters: *"It would be so great if we got our job-critical data from the same platform so that we all get accurate and necessary information anytime and anywhere"* [P7].

Participants were painfully aware of numerous open challenges (or stubborn problems) and attempted to reconcile this with the ubiquitous presence of technologies in the nutrition arena. The whole process requires many resources, including manpower, skills, and time, and great effort to accomplish. The popularity of paper-based surveys over digital data collection is increasing dramatically in the development sectors. However, while ICT has started using data collection through tablets and mobile phones along with the traditional method (paper-based), there is a wider scope of innovations to explore in these areas to work towards user-friendly M&E systems. A specific challenge was found in the M&E systems that every NGO and government organizations carried out within their programs: *"We need to find a better way to do monitoring ... the way we do it involves a lot of labor including data collection, managing many indicators. I think it's too heavy, we could change it, and we need more innovation here"* [P6].

We noticed varied levels of understanding of the feasibility of technology-led solutions, and the "innovate at all costs" agenda often motivated a number of participants to ignore the applicability of digital solutions within existing national technology infrastructures. The participants also noted the need for quick and auto-generated information using DT to access essential data and indicators. Complex field realities also risk being squeezed into narrow technological funnels to appease decision-makers' desire for digestible information flows: *"very recently we have developed a real-time monitoring system which can automatically monitor how much progress we have made in the field"* [P2]. In this instance, the underlying assumption is that through the introduction of digital M&E systems, decision-makers based at headquarters would have easy real-time access to data collected by field extension workers.

Finally, another aspect of interviewees' views and expectations of ICT innovations is that most of them seem to have an imaginary sort of perception that innovation magically tailors technology according to their challenges and needs in service delivery and program management. Many participants saw digital innovation as a way to automatically generate simple solutions and alternative approaches to tackle complex challenges in their jobs. For instance, a policymaker interviewee shared his belief that technology could bring a new solution to handle particular challenges in delivering information to low-literacy marginalized groups: *"we would like to transfer expert knowledge to the marginalized farmers. I think technology can do something in this area. I don't know how but I know there are some algorithms by which the software engineers can easily build an app which can be able to transfer complex knowledge in an easy way to the farmers"* [P8]. This view indicates how top-level decision-makers operate with a partial understanding of the possibilities of digital technology, in part due to limited investment in learning about the opportunities and challenges of digital innovation. The reality is even basic expectations around technology infrastructures (e.g., electricity availability, cellular network reach) are not reliable assumptions to make in development contexts [43]. A government official stated that although it was impossible to provide digital resources to communities located in areas without appropriate digital infrastructure, an innovation that worked without these underlying infrastructures could be deployed, although it was not deemed necessary to consider how this might be done. *"People who live in remote areas* *can't access the internet, and we face problems in providing digital-based training to them. It would be really good if we had innovative ideas that can be done in areas without internet and if possible, without electricity!"* [P1]. In such instances, technology innovation is viewed as one-size-fits-all and the inherent difficulties in reaching rural areas with technological solutions are minimized.

In summary, our decision-making stakeholders shared that they look for new approaches and innovations to improve quality, service delivery, effective work management with human resources, efficiency, and regular M&E of their interventions and programs. Here, innovations are often shaped by their accountability to their higher-level stakeholders. However, among the government decision-makers, there are politically-motivated tendencies to achieve national recognition by association with innovations. Our findings also suggest that decision-makers in nutrition program development often have unrealistic expectations of innovations. This kind of fixed-mindset preconception of innovation indicates that top-down decision-makers often lack understanding of limitations of feasibility, applicability, and technological advancement of the population served. The promise of technology in the sector and its associated political incentives also create an environment in which stakeholders have unreasonable expectations of technology.

5.1.4 Default Association of Young People with Technological Adoption. The youth constitutes more than a third of the Bangladeshi population [63]. The Bangladeshi government is committed to making digitization a national priority by increasing focus on youth engagement in technology-based programs and projects. As one NGO interviewee noted, *"part of the election manifesto this year is to target youth groups to engage with technology for youth skill development and empowerment"* [P10]. The government has also called for more effective skill development among the youth, especially in the technology space, to ensure their employability in diverse industries. Our participants echoed this government's new priority and shared their perceptions on bringing more youth-centered development programs. Youth are often considered natural technology users and adopters, and their enthusiasm and curiosity for early technology adoption are seen as strategic considerations to focus on in technology programs. For instance, one UN official noted the emphasis put on youth as quick learners: *"they can learn the technology so quickly, and they are the future of the generation, so it's worth focusing on young people!"* [P6].

While there is a common tendency to create youth-centered ICT programs, however, there is no bottom-up program design approach that includes design consideration for youth-centered ICT initiatives which are appropriate for them. Such initiatives can pose threats to the privacy and safety of vulnerable people. The issue of misuse of technology by youth has also been noted by our participants: *"Young people are motivated to use technology, but at the same time we are concerned about inviting trouble…so many risks, particularly for girls and women"* [P14]. While effusive about the potential for technologies to transform the development landscape, decision-makers also expressed reservations about technology use, influenced by their previous experiences of using technology in programs where such issues emerged: *"I have seen the troubles that technology brings like abuse, privacy breaches, and security issues especially for girls and women"* [P10].

While young people have been seen as central to Bangladesh's future, there has been less focus on other groups, especially the elderly population, who are often ignored within ICTD programs: *"The youth group is critical. We don't put too much hope in elderly people; they have already passed half of their lives, so the new generation should be the focus for our efforts"* [P1]. Hence, focusing on one specific age group raises questions on how much ICTD ensures inclusive technology for different ages and levels of marginalized groups (such as marginalized women and elderly people in Bangladesh) and on what measures are being considered to tailor ICT initiatives for youth, which need to be carried out with the utmost care (by engaging them during design technologies and programs).

In summary, there is a common tendency to link young people to any kind of innovation without proper consideration of youth-centered design. Further, there is a lack of provision for other groups, such as ICT-based health and nutrition interventions for the elderly population.

5.2 Utilizing Evidence-Based Approaches: Adoption, Adaptation, and Strategic Engagement for Scalable and Sustainable Program Development

Our participants' approaches to evidence-based policy and program creation placed a particular emphasis on learning from neighboring countries' experiences and leading global countries in developing nutrition programs. *"During our dairy policy formulation, we followed the policy of India, Sri Lanka, and other similar neighboring countries. We have also compared African countries' policies, as we have similar economic conditions"* [P3]. Hence, there is a common tendency among decision-makers to adopt the same interventions and approaches that have been used in other global contexts to address similar problems in Bangladesh. In a typical scenario, based on their efficacy analyses from other countries' experiences, successful programs are adopted after making contextual adaptations. For example, one participant outlined the need to adapt a worthwhile U.S. project for Bangladesh, where they needed solutions that *"engaged young people in nutrition education through technology. One approach ... originally from America, but today many countries are also following this ... we are thinking about and analyzing how this 4H model*[18] *[4-H pledge: head, heart, hands, and health] for engaging youth can be adapted in Bangladesh."* [P10].

Participants emphasized the importance of assessing a program's relevance to determine its potential feasibility and how it measured up against national development goals. One of our policymaker participants noted the questions they had while reviewing the scalability of innovative approaches they encounter: *"What will be the outcome? Is it making any changes? How much is it changing and how easy is it to implement? Is it a people-friendly initiative or not? How much effort do you need to put in to increase its efficacy?"* [P2]. Participants also noted that they worked in a rapidly evolving development context, which meant that they regularly needed to *"check the current nutrition status of the marginalized population such as women and children, and then look to the international-level evidence for comparison"* [P4]. However, it is important to note that these evidence-based statuses are based mainly on national-level health and nutrition surveys, which often consist of quantitative data and do not derive from marginalized communities' views of their actual problems. Government surveys and reporting systems do not capture citizens' lived experiences and challenges in the most meaningful way-in their own voice.

Sustainability has been seen as an important criterion when designing initiatives. One policymaker participant commented, *"short-term innovation projects and pilots are not the solutions of a certain problem. It has to be a sustainable plan for the long term"* [P1]. The additional work that programs require to be feasible changes from program to program. While sustainable program development is their core requirement, all of the participants mentioned that community participation happens during the program implementation stage and not during the design phase. They elaborated that sustainable program development considers careful adjustment and local adaptation to ensure a more robust and durable scale-up. In many instances, *"for the technologies to be socially acceptable, we need to do many women empowerment activities in the field. We develop awareness-building exercises with women's households to get other members in their families, such as their husbands and mother-in-law, to sustain the technology. We do these at the community level"* [P11].

One of our NGO interviewees shared their experience of modifying a particular nutrition program during the implementation stage. He shared their investigation on how changing their focus

[18] https://4-h.org/

from one group (university students) to another group (school students) helped them scale the "nutrition club" programs across Bangladesh. He stated: *"We realized that schools are the better option for establishing nutrition clubs ... we moved our main focus for nutrition club from university to school students for the program's sustainability"* [P10]. It is important to note that our participants mentioned the importance of an evidence-based approach based on learning from other countries, policy documents, surveys, and other global initiatives to adapt based on efficacy and relevance analysis for program and policy development.

However, none of them mentioned the bottom-up approach for program and policy design, incorporating communities' actual needs, challenges, priorities, and capabilities for sustainable program and policy development. We also found another approach to sharing evidence of NGOs' ongoing programs with policymakers. Participants mentioned that direct interaction between policymakers and program implementers from development organizations is crucial to informing decision-makers about their evidence-based innovations and initiatives. When engaging with decision-makers, development NGOs (especially the most influential ones) make concerted efforts to involve government policymakers from the earliest stages of program design and implementation. This is development realpolitik at play, as NGO leaders seek to give policymakers a sense of ownership in the success of programs they carry out.

Participants described positive results that emerged from their decision to involve policymakers in their programs, which in turn gave them opportunities to influence government policies and programs. One participant noted that they *"involved policymakers from the very beginning, invited them to our workshops, and asked them for feedback and recommendations on our initiatives in hopes for opportunities for policy uptake and scale-up"* [P10]. Participants emphasized that repeated and continuous involvement of policymakers in their programs have positive effects on government policies and programs. We saw another example of this phenomenon with P10, who is the CEO of an NGO that carries out digital empowerment and nutritional awareness programs. He discussed how a strategic decision by his NGO to involve senior government officials (for instance, through speaking engagements at gift-giving ceremonies for NGO programs) resulted in opportunities to share their nutrition programs with important government ministers. One of their key community nutrition initiatives has now become a government-sponsored national program that is carried out every year: *"We invite top policymakers from the Ministry of Agriculture, Food, Health, and Technology ... We engage them in our innovative initiatives [technology-based interventions] and connect them with youth groups from nutrition clubs"* [P10].

In summary, at the top level of the decision-making context, there is a common tendency to adopt interventions and approaches from global and neighbor countries' contexts. However, some important criteria for nutrition-program development are relevancy to the contexts, evidence from national surveys and reports, feasibility and acceptability from communities and sustainability approaches. Our participants shared some political intentions for creating evidence-based approaches for nutrition program development through strategic interactions. These strategic interactions often create direct and regular engagements with policymakers that lead to a development program being scaled up. Hence, our findings suggest that strategic interactions-direct and regular engagement with policymakers-can significantly impact a program scale-up at the national level. Finally, we found that there is a lack of consideration of a bottom-up approach for program and policy design.

5.3 Understanding Multi-Sectoral Stakeholder Collaborations: Competition and Power Dynamics

Stakeholder collaboration in international development is complicated, as a large and diverse set of actors are involved in decision-making. In Bangladesh, 24 distinct multi-government ministries, along with many large international NGOs, UN agencies, research organizations and donors, operate within the nutrition policy space. The global movement for achieving food and nutrition security through SDG 2 highlighted the need for the development of nutrition-sensitive agriculture food systems through multi-sectoral stakeholder coordination. Therefore, in light of the sheer scale of national nutrition policy that is carried out in this nation of more than 160 million people, development agencies seek to value each other's contributions. Indeed, our interviewees commented that fostering connections with other agencies who work within their space is a top priority for them. However, while the interest in doing work together in particular programs and projects is real, participants noted that real examples of collaboration are weak: *"Some NGOs have good project materials, but they don't know dissemination strategies, and others have the physical infrastructures ... at the end of the day, all are isolated"* [P10].

It is difficult enough to collaborate across the divide between the public and third sectors, but according to some of our interviewees, even intra-sector collaborations are difficult. One senior UN official remarked that *"it's complicated; the Ministry of Agriculture does not want to hear from the Ministry of Health, and the Ministry of Health does not want to hear from the Ministry of Education. So, there are so many problems and not many strong relationships. There should be good coordination, but it seems like coordination among them is still very fragile"* [P7]. Hierarchical power dynamics among stakeholders is a well-known reason for weak collaboration that results in tension and poor coordination among the stakeholders. Collaboration among organizations at similar levels is complicated, as it is shaped by competition and ownership issues, such as who will get specific funds or, lead a specific program, and thereby become more prominent than another. Participants revealed that the agencies that ought to be working together most closely were often in competition with one another. The ensuing frustration from this weak collaborative environment was evident when we spoke to an NGO worker who stated: *"officially, nutrition [programs and projects] is mandated by the Ministry of Health, and food safety [programs and projects] is going to be under the Ministry of Agriculture's responsibility, although both are connected. But there are challenges about ownership, like which ministry will direct what program?"* [P14]. In general, reaching top-level decision-makers such as at the Ministry of Health and Agriculture was the most challenging for local organizations. One participant from an NGO saw working with them as strenuous: *"It's tough to convince the Ministry of Health and Agriculture. They are like a huge titan. It takes lots of effort and time to work with them"* [P10].

Reasons for the disjointed relationships include a lack of clarity about other agencies' objectives, unclear division of responsibilities, and a lack of a perceived need to strategize together. In addition, one interviewee opined that *"the teamwork mentality is not there yet ... their focus is more on their own priorities for the programs. Moreover, collaboration is a matter which reflects during the implementation stage by matching the interest with each other's own priorities and goals which creates more complex situations for us to collaborate"* [P1]. Although the government publishes regular reports on how development outcomes are being met, questions such as "who will work together on particular outcomes?" or "how will the work be allocated for maximum effectiveness?" are not addressed at the national planning level. However, our participants confirmed that stakeholder coordination at the planning stage is promising. Many interviewees expressed a desire for stronger inter-agency collaboration, suggesting that they could work with other government departments and NGOs in activities such as program document creation (such as evaluation reports and strategy presentations) to share with top-level government ministers and donors. This motivation for partnerships is also politically expedient. A UN official articulated why there is a need to work together: *"We are good at producing lots of documents and reports, and everyone is interested in these because it links with donors' interests. [There are] lots of good documents but taking these to the implementation stage is the problem"* [P6].

Finally, although the nutritional development space is a crowded marketplace, some participants argued that key organizations from the private sector are missing from discussions altogether: *"If they could bring the private sectors, they should become more responsible. If you kick them aside, they will not listen to you, or you cannot influence them, rather let them get on board and influence them"* [P11]. Many interviewees also noted that field workers and decision-makers from rural communities were underrepresented within existing collaboration structures. NGO workers noted that such under representation creates challenges for implementation: *"The field-level people involved in implementing projects at the ground level are absent; only the top-level people were involved in policy, which actually made the policy complex and complicated to implement"* [P8].

In summary, while multi-sectoral stakeholder collaborations are crucial in nutrition policies and programs in Bangladesh, they are in practice still weak and disjointed, especially in intra-sectoral collaborations among similar-level organizations due to lack of teamwork mentality; clarity about each other's objectives, responsibilities, and priorities; and internal competition for lead programs. However, our participants noted that stakeholder collaborations are positive in planning and document creation (especially in M&E and strategy presentation) stages, although there is often a political motivation linked to the highest levels' actors, such as donors. Participants also highlighted the limited presence of bottom-level actors in the multi-institutional decision-making stakeholder contexts in nutrition.

6 DISCUSSION

In our findings, we have described the priorities, experiences, and expectations for innovation in the development and implementation of Bangladesh's national nutrition programs as noted by our participants. We have further presented how global agendas, political priorities, hierarchical power dynamics, and donor interests shape ICTD programs and impede bottom-up approaches by limiting community participation. Our study highlights digital innovation for nutrition development in Bangladesh is detached from Bangladesh's digital vision. We have also reported the complex power dynamics among diverse decision-making stakeholders (specifically among similar-level organizations) that create tensions and competition in program development and implementation. Thus, our findings reinforce previous findings from the CSCW and other literature that working with diverse decision-making stakeholders is complicated and difficult, specifically within the dynamics of government, NGOs, and donors in international development. Our findings highlight the challenges of working within Bangladesh and the Global South, and raise specific considerations for how to configure CSCW and HCI work within the development sector in this context. In the following sections, we discuss these considerations for sustainable innovation, linking them to policy-making and how to leverage innovation diffusers for broader groups of citizens, and the key implications for social computing and HCI4D researchers working in ICT-based program and project development. Although based on considerations of public health nutrition in Bangladesh, we discuss how these findings can be generalized to similar development contexts.

6.1 Understanding Decision-Making Contexts and Positioning for HCI Researchers

Our findings show how the priorities of higher-level stakeholders, such as governments, UN groups, and donor agencies, shape ICT innovations in nutrition development programs in Bangladesh and that the opportunities for bottom-up approaches to innovation in public health nutrition in Bangladesh are still limited. The tendency of governments and donors to push technology innovations into Global South contexts is not a new phenomenon, as discussed in literature that we highlighted in our related work section. However, distinctions between the ways in which top-down authority is invoked is necessary to clarify that the actions of high-level policymakers, government officeholders, and multinational NGOs are not necessarily antithetical to the success of development innovations. Those involved in CSCW and HCI4D have long advocated for engaging with under-represented groups and bringing under-explored topics to the fore. We extend this line of work by arguing for incorporating marginalized communities' voices as part of an engagement with program and policy makers into the decision-making process. Our findings highlight that this incorporation is often challenged by various internal and external factors in the local governing system. By considering the interactions among diverse top-down institutions [58, 76, 114, 115] their institutioning, the intermediaries involved (such as multi-sectoral actors in government INGOs and donor agencies), and the dynamics of intermediation, we can articulate how they work and where HCI can contribute in the processes of digital innovations with and for marginalized communities. Hence, we suggest that HCI researchers develop a nuanced understanding of the policy and program-making contexts within which their intervention is placed. We suggest there is a potentially fruitful opportunity for HCI researchers to take a middle-man approach and connect the bottom-up and top-down contexts [37].

Intermediaries already exist in institutions' systems (e.g., policymakers) to improve program implementation and sustainability. We argue that, while these intermediaries exist, HCI researchers can also play the role of a middle-man or mediator to make the bridge stronger within both top-down and bottom-up stakeholders. We propose that HCI researchers expand their roles and move into a as mediator or middle-man role to liaise between decision-makers and bottom-level users. This strategy can bring the designer closer to understanding the greater context of implementation through improved methodology to work for sustainable ICT development in the Global South with a more nuanced and context- specific understanding that allows for potential sustainable design. We propose four dimensions of mediation (external, internal, vertical, and horizontal) that can be applied to better engage with high-level decision-makers. We suggest that understanding factors in terms of these four dimensions will help to better understand the decision-making context.

First, we need to consider **external** factors and actors. More specifically, CSCW and HCI4D designers and researchers need to know who the prominent multi-sectoral decision-making institutes and actors are (such as governments, NGOs, UN groups, and donors) in a targeted international development sector (such as public health nutrition). We need to be aware of and consider all external influences and motivations where these decision-makers work as "innovation intermediaries" [111] to shape and implement innovations. For instance, within these diverse top-down institutions, CSCW and HCI researchers need to consider external factors of institutioning [76], such as how the global and national mandates, agendas, evidence-based approaches, and priorities constantly move and shape innovations. Second, we need to consider **internal** factors to understand the dynamics of these intermediaries [58, 114, 115] by understanding their political motivations, existing competitions, perceptions, and expectations of using ICTs in development programs and policies. Third, we need to understand **vertical** mechanisms and interactions. For instance, we need to consider mechanisms such as the way top-down approaches and actors such as decision-makers inform the bottom-up actors such as marginalized citizens and vice-versa [38]. Finally, we need to consider **horizontal** mechanisms and interactions by considering existing practices and processes within innovation. These include understanding how these diverse decision-making actors work and collaborate with each other and the dynamics of their diverse collaborations and intermediaries. Consideration of both internal and external factors are also needed within both vertical and horizontal mechanisms of interaction. Accounting for the relevant dynamics in terms of these four dimensions can help HCI and CSCW researchers to understand the top-down decision-making environment for innovations and discover the gaps that need to be filled to successfully implement bottom-up approaches.

6.2 Configuring for Sustained Impact

As we have highlighted, there is an opportunity for HCI researchers to engage as mediators to foster discussion around bottom-up approaches to technology innovation. In such a role, researchers would be in a position where they can both understand and influence the design of specific interventions for sustained impact. By engaging as a mediator between the decision-making and community contexts, there is the potential to advocate for incorporating marginalized communities' voices into the decision-making process. Spaa *et al.* [109] specifically emphasized that HCI researchers' role in the top-down context is in capturing citizen voices, and more action is needed to incorporate them in the decision-making process. We argue that while designing for a particular community or situating a work within a hyper-local context, the broader context of development needs to be considered. This broader international development context includes the work that has been carried out for decades by local development NGOs, international NGOs, donors, and local governing bodies. Involving these stakeholders will enable HCI researchers to target priority development outcomes both locally and regionally.

Understandably, many HCI4D researchers design intervention in such way to minimize complexity. Choosing not to address difficult questions about politics (confident that their local project partners will engage with such issues if necessary), they finish their research, and then ask, "where do we go from here?" In many cases, the questions of scale, long-term sustainability, and advocacy for policy change are invoked as afterthoughts, perhaps because these are considered areas of expertise of our local project partners rather than ourselves, or the intervention gained specific (or unexpected) traction when deployed. We suggest that these questions should be proactively considered from the very beginning of a project, in dialogue with relevant stakeholders. For instance, when designing for diverse populations from small villages, it might be useful to start conversations about the study with (and potentially involving) local decision-makers, e.g., municipal officers and extension workers from government ministries who design strategies for the local region. Undoubtedly, this introduces the potential for delays and additional complexity to the project, which would have otherwise been ignored. Nevertheless, an approach that engages with messy realities on the ground is more likely to lead to long-term impact. Hence, it is crucial to engage with both top-down and bottom-up approaches to technology design and interventions and avoid a dichotomous view that pits one against the other [55]. Policymakers also need to understand the nuances of the impact and popular appeal of technological platforms, and we posit that HCI scholars need to play an active role in the education of our partner organizations through engagement and conversation [70]. Indeed, our findings emphasize that there is a need for a strategic plan to engage policymakers from the very beginning of the project. A community-led, bottom-up approach can thus be strengthened through regular informing of decision-makers to help them better understand the feasibility and accessibility of the innovation for the broader sustainable program and policy landscape.

Taking a closer look at some examples of technology innovations that have been able to create sustainable impact, we see that in addition to a commitment to work within local constraints and infrastructures, practitioners emphasize the value of collaborations with multiple NGO stakeholders and/or government agencies in contributing to their impact. As an example, Ushahidi (a platform for local communities to share their stories) [82, 102] was successful due to strong collaborations with local NGOs, which not only added a layer of trust to NGO-verified contributions but also increased uptake [119]. Similarly, the 'Our Story' (currently known as 'Indaba') participatory video platform was pioneered by HCI researchers by leveraging the institutional networks of the Red Cross Movement's umbrella body IFRC (International Federation of Red Cross and Red Crescent Societies, who operate on both top-down and bottom-up dimensions [126]). Through this collaboration, a number of relationships with NGOs at the local level were brokered (with the Indonesian Red Cross and Namibian Red Cross, for example) [11] which were used by the IFRC to advocate for sustained use of the tool within the wider network. Finally, RootIO (a platform for low-cost community radio) similarly credits their success to extensive collaborations with content partners, politicians, community leaders, networks of youth centers, and contacts within government ministries [26, 29]. Indeed, RootIO collaborated with national communications regulators and involved local community NGOs [28] in each deployment. We recommend that as part of the requirement of initial information-gathering within social computing and HCI for development project collaborations, researchers should find out the priorities of local political stakeholders who play influential roles in regional development and areas of alignment or difference between collaborators and other NGOs and government ministries who work in that sector.

Moreover, decades of work in Participatory Design (PD) has been advocating for such inclusion of all stakeholders; in fact, there have been a plethora of examples in HCI of local government bodies being included in designing projects of public interest. Through configuring the role of intermediation and understanding how an artefact is shaped, embraced, and used in the real world, PD in HCI can improve its approach to effective social innovations [25]. We want to emphasize that we advocate taking this strategy further. We do not only mean having some policymakers around the table along with other stakeholders to make the design more democratic, but also that they need to design for the challenges that policymakers face and involve local people in designing solutions for those problems. For example, as our data shows, policymakers are often tempted to deploy digital applications because there is external pressure from donors and the government but that have little use in the community. We suggest that this problem also be brought to the table when doing such as designing a nutrition program with the local community. A grassroots-level movement, for example, may create a nationwide awareness against such unjustified pressure to deploy digital technologies. We are excited about the potential of various HCI interventions that have allowed local people to voice such concerns in the Global South [11, 81, 84, 121, 122], and we believe that such technologies might be useful in addressing the problems at the top to help make a bottom-up design successful.

We also recommend exploring potential design interventions to include community voices for ICT-related program development and implementation. Our findings reinforce the fact that, thus far, community participation occurs (if at all) *after* program development and there are limited connections to bridge the gaps between decision-makers and bottom-level marginalized communities who need to share their needs, challenges, and priorities so that an appropriate program can be tailored for them. We would also like to raise critical research questions for future experiments. We argue that more qualitative research is required to understand the challenges decision-makers face in incorporating the community's views in the decision-making process before policies are implemented. Based on such knowledge, CSCW and HCI4D scholars should explore how design can support innovation. For instance, how can community voices be incorporated in calls for proposals (also known as "calls for expressions of interests", "calls for tenders," and "calls for bids") for program development? We argue that community voices should be incorporated in calls for proposals, as this is one of the most important pre-stage of any program and project development in international development contexts. To our knowledge, this is an underexplored space for ICTD research that provides opportunities for innovation and sustained impact.

6.3 Tech Obsession in Policymaking

Our findings show that there is intense interests in developing new technology-related innovations among policy and program creators. We raise this as a critical challenge for ICT and HCI researchers, as this tendency to create new technologies raises a tension between their central priorities of "innovations" and "sustainability." For instance, while policy and program makers have been also pursuing development of novel technology innovations for program management, our participants have advocated the need for more evidence-based sustainable program development. This intention create additional new technologies is a threat where questions of sustainability and scalability of those technologies are substantive, leading to many instances of technological innovations that fall under the heading of "solutionism." This tendency is often cited in social and political science critiques as "technology solutionism" or the "technology fix." In recent HCI design research, fixing problems that don't exist and proposing"quick fixes" or "silver bullets" for complicated social, political, and environmental challenges are criticized as "solutionist" [18, 19, 75, 86]. The "techno-solutionism" ideology was critiqued by Morozov [85], who pointed out the political tendency to believe blindly in the ability of technology to solve challenges with innovative design. On democratizing the potential of the internet and ICT, Morozov also highlighted a critical point of debate on political transformation in digital societies, expressing skepticism toward governments' ability to promote democracy through the internet in the West, adding that "techno-solutionism" ignores debates about politics, history, culture, actual needs and challenges, and people's capacities to adopt new technologies [85].

Such tendencies can be seen as a trend towards "magical thinking" regarding technology and institutional systems, which can be seen as a historical pattern. Larry Cuba pointed out that this tendency has a political aspect that goes beyond social factors and triggers political motivations [30]. These concepts bear resemblance to the concept of the "sociotechnical imaginary" that is broadly used in the field of social technology. Sociotechnical imaginaries are generally future-oriented visions of connected social and technological orders operated by key individuals and institutions [64]. Although scholars have noted that sociotechnical imaginaries can identify paths and actions for effective innovations, they caution that disparate imaginaries can bring very different programs and policies to society [64, 108]. Our study also contributes to this growing area of literature on the relationship between decision-makers' power and techno-scientific innovations. As social computing and HCI4D researchers, we want to raise the possibility of positions and initiatives that address this techno-focused solutionism among decision-makers so as to resolve the contradictory vision between innovation and sustainability. Indeed, such tech obsession has already been reported in the CSCW and HCI4D literature (for example, see [83]), and various critical and speculative design interventions and ethnographic studies have been done in the HCI literature to counter such utopian thinking [10, 36, 97, 106]. However, little has been done to operationalize this criticism in practice that can significantly impact policymaking. Moreover, a deliberate effort towards affordable (in both financial and human resource) technology design is required, using existing platforms to capture community voices in ways such as using participatory media to invite marginalized voices and to share with decision-makers to inform programs and policies. Such efforts should consider designing systems according to both high-level and bottom-up needs and capabilities. Additionally, as suggested by our findings, instead of building more social-media-style platforms, un-platforming innovations that leverage existing social media channels offer an alternative route to engagement. Un-platformed design is a conceptualization of coordinated participation in social media technologies, [71]. This study provides three exemplar projects in participatory health research, connective online learning, and strategic future foresight through the configuration of accessible and well-known social media apps such as WhatsApp and Facebook that act as clear examples of how this could be used. We also emphasize to social computing and HCI4D researchers that by building on existing platforms, practitioners could avoid the pitfalls of developing expensive bespoke solutions that may or may not work.

Our participants had many unrealistic expectations of digital innovation, indicating lack of understanding regarding the uses of technology and its feasibility among decision-makers. Our findings reflect the work of Gurumurthy [48], who claimed that when thinking about innovation, without reasonable understanding and expectations of what technology can and cannot do, the wicked problems of international development will only become more wicked. Enhancing digital literacy for the marginalized population has long been well-recognized. From our interviews, we have found that the tendency towards techno-focused solutionism reflects a lack of awareness of the negative impacts of technology among decision-makers. We have also identified that, notably, while there are diverse multi-sectoral decision-makers present at the top-down level, there are no ICT and HCI designer positions at that decision-making level. This was found to be one of the reasons for the lack of understanding about the potential negative impacts of technology deployment. We believe, and want to emphasize, that including critical HCI scholars and practitioners in policymaking teams (as mediators) is essential to reduce this bias.

6.4 Leveraging Innovation Diffusers

Sustainable impact requires interventions to "recognize and leverage resources already existing in the community, including physical and social infrastructure, skills, knowledge, networks, and environmental resources" [95]. Partly driven by a political mandate to engage youth explicitly as part of their digital innovation strategy, our participants argued for leveraging youth potential, seeing them as agents of change. This a priori interest in youth engagement signifies how decision-makers think of population segments, which will be key in successful innovation diffusion. Notably, this is not limited only to creating youth-friendly applications or creating "lite" youth versions of apps aimed at adults. Youth are traditionally construed as "human becomings," "denigrating their present lives, while focusing only on the adults they will become" [7]. For decision-makers, the youth of today have an essential role to play in Bangladesh's digital innovation narrative in the present. This is also a strategic decision when it is considered that 40% of the country's population is under the age of 18, a phenomenon that is not exclusive to Bangladesh [104]. The UK's leading development institute, ODI, argues that since the world now has the largest generation aged 15–24 in history (and 90% of these youth live in Global South countries), a unique opportunity is afforded to development practitioners and theorists. There is "a one-time window of opportunity for a concerted international effort to help developing countries reap a 'demographic dividend' from educated, healthy and gainfully employed young women and men, and achieve far higher economic growth rates" [96]. However, we also urge caution here and advise that any attempts to mediate youth engagement with decision-makers needs to be carried out with the utmost care. Though ostensibly motivated by youth empowerment, authorities as well as non-state actors have repeatedly sought to disrupt spaces for youth participation or co-opt them towards their own political agendas. Moreover, if development projects consider youth perspectives and motivations for development, they can be included in initiatives whose primary target populations are adults. For instance, when targeting information dissemination to women farmers, we should consider young people's role as change-makers in farming communities. Hence, we recommend that every national and regional development context has population segments that are prioritized for innovation roll-out due to factors such as high early adoption rates or alignment with national agendas. When configuring projects, we recommend that HCI researchers consider how their projects will leverage these population segments to maximize their impact.

6.5 Limitations and Future Work

We must acknowledge some limitations to the generalizability of our study. Our study only engaged a group of decision-makers who work in the nutrition development context, which might differ from other development sectors in the country. While we believe that most of our findings may hold for a broad range of public health nutrition settings in Bangladesh, we acknowledge that the findings might have been different in other settings. Hence, we refrain from any kind of generalization of our findings beyond the studied setting. Moreover, we acknowledged other scholars' contributions broadly in our related work and discussion sections who raised many similar issues in broader ICT for development contexts. However, one of our unique contributions to the knowledge is we have taken a specific development setting, namely public health nutrition under the digital health umbrella, and deeply explored critical factors with evidence from our findings. To our knowledge, this is the first study to engage policy and program decision makers in public health nutrition development context to explore their motivations, experiences, expectations, and current multi-stakeholder collaborations in digital innovation to understand how these perspectives broadly impact sustainable ICT development in Bangladesh. To take necessary steps toward incorporating the needs and priorities of the bottom levels into high-level decision-making, we suggest additional research in other development settings such as agriculture, education and climate change to provide a broader picture for innovation development research in the Global South. We have also raised several questions in areas such as the challenges decision-makers currently face in incorporating communities' views in the decision-making process and on how social computing design can contribute to scaffolding real-world challenges towards sustainable ICTD. These need to be addressed by taking in-depth qualitative research initiatives to understand the underlying factors involved.

7 CONCLUSION

Achieving sustainable impact is a crucial concern within digital interventions in international development. We explored this challenge by taking a closer look at Bangladesh's nutrition development context and the way influential decision-makers (i.e. government officials and significant NGO leaders) approach digital technologies. Our findings show how high-level decision-makers in this context prioritize incorporating innovations in programs and their perceptions and expectations of innovations in complex, diverse decision-making stakeholder collaborations. Our findings suggest that disjointed collaboration among decision-makers leads to unnecessary duplication of efforts and unrealistic expectations of digital innovations, which impedes sustainable ICTD from designing contextually appropriate public health nutrition solutions in Bangladesh. We highlight the typical decision-making process for proper innovation program development as highly polarized by high-level stakeholders, which continues to limit bottom-up approaches. Finally, to move forward, we propose four dimensions (internal, external, vertical and horizontal) in which to understand top-down decision-making contexts as a route to configure research for sustainable impact. Taken together, we suggest for social computing and HCI-for-development researchers that bottom-up development needs to be approached with a deeper understanding of decision-makers' interests and challenges. To do that, we recommend, CSCW and HCI researchers become program design mediators to connect decision-makers and communities, bring marginalized community voices to ICT-based program development, consider careful innovation designs for youth, and make efforts to leverage existing technologies and platforms for sustainable ICT innovations for global development.

ACKNOWLEDGMENTS

This work would not be possible without the generosity of our participants. We are deeply grateful to our research participants who made the time and effort to give us valuable and detailed responses.

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Received January 2021; revised July 2021; accepted November 2021